Qumulo Nexus Configuration Guide



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Authentication in Qumulo Nexus

Configuration Workflow for Storage Administrators who Manage a Qumulo Nexus Account

This configuration workflow explains how to enable single sign-on (SSO) for Qumulo Nexus from the perspective of a storage administrator who manages a Qumulo Nexus account and works together with a system administrator who manages your organization's identity provider (IdP).

After the storage administrator performs the initial configuration in Nexus, the system administrator configures the IdP. Then, the storage administrator completes the SSO configuration in Nexus.

Prerequisites

- Administrative access to your organization’s Nexus account
- A subdomain for your organization

Step 1: Perform Initial Single Sign-On (SSO) Configuration in Qumulo Nexus

Before you ask a system administrator to configure the identity provider (IdP), you must perform the initial SSO configuration in Qumulo Nexus.

To Perform Initial SSO Configuration in Nexus

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On your organization’s page, click **SAML SSO**, and then click **Configure SSO**.
4. On the **Configure SAML SSO page**, enter a **Nexus login subdomain** and then click **Save & Continue**.

   The SP Entity ID (your Nexus account’s unique identifier) and ACS URL (the Assertion Consumer Service URL that receives SAML responses) are displayed. For example:

   ```
   https://mysubdomain.nexus.qumulo.com
   https://mysubdomain.nexus.qumulo.com/api/v1/auth/saml/acs/
   ```

> **Note**
>
> - Record these values for the next step.
> - If the subdomain that you want to use is unavailable, choose another subdomain or contact the Qumulo Care Team.

Step 2: Ask a System Administrator to Configure an Identity Provider (IdP) for Qumulo Nexus

After you perform the initial SSO configuration in Nexus, you must ask a system administrator in your organization to configure the IdP and then provide you with the IdP Metadata URL.

1. Provide your system administrator with the SP Entity ID (your Nexus account's unique identifier) and ACS URL (the Assertion Consumer Service URL that receives SAML responses).
2. Ask the system administrator to perform the necessary configuration (page 6).
3. Receive the IdP Metadata URL from your system administrator. For example:

```
https://my-organization.idp-provider.com/app/abcd12e345fgHIJKLm678/sso/saml/metadata
```

> **Note**
> The format of the IdP Metadata URL depends on your organization's IdP provider.

Step 3: Perform Final Single Sign-On (SSO) Configuration in Qumulo Nexus

After your system administrator configures the identity provider (IdP), you must perform the final SSO configuration in Qumulo Nexus by using the IdP Metadata URL .

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On your organization's page, click **SAML SSO**, and then click **Configure SSO**.
4. On the **Configure SAML SSO**, enter the the **IdP Metadata URL** and then click **Complete Configuration**.

   SAML SSO - Enabled is displayed.

Next Steps

After you perform the final SSO configuration, you can click **Users** and then add users to your Nexus account. Every user that you add has SSO enabled by default.

Configuration Workflow for System Administrators who Manage a Qumulo Nexus Account and an Identity Provider (IdP)

This configuration workflow explains how to enable single sign-on (SSO) for Qumulo Nexus from the perspective of a system administrator who manages both a Qumulo Nexus account and an Identity Provider (IdP).

After the system administrator performs the initial configuration in Nexus, she must configure the IdP. Then, she can complete the SSO configuration in Nexus.

Prerequisites

- Administrative access to your organization's Nexus account
- A subdomain for your organization

Step 1: Perform Initial Single Sign-On (SSO) Configuration in Qumulo Nexus

Before you can configure the identity provider (IdP), you must perform the initial SSO configuration in Qumulo Nexus.

To Perform Initial SSO Configuration in Nexus

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On your organization's page, click **SAML SSO**, and then click **Configure SSO**.
4. On the **Configure SAML SSO page**, enter a **Nexus login subdomain** and then click **Save & Continue**.

   The SP Entity ID (your Nexus account's unique identifier) and ACS URL (the Assertion Consumer Service URL that receives SAML responses) are displayed. For example:

   ```
   https://mysubdomain.nexus.qumulo.com
   https://mysubdomain.nexus.qumulo.com/api/v1/auth/saml/acs/
   ```

> **Note**
>
> - Record these values and provide them to the system administrator who manages your organization's IdP.
> - If the subdomain that you want to use is unavailable, choose another subdomain or contact the Qumulo Care Team.

Step 2: Configure an Identity Provider (IdP) for Qumulo Nexus

After you perform the initial SSO configuration in Nexus, you must configure your IdP.

> **Tip**
> If you work together with a storage administrator who manages your organization's Nexus account, she provides you with the SP Entity ID and ACS URL.

To Configure Your IdP for Nexus

1. Log in to your IdP's console.
2. In the application or service configuration section, take the following steps:
    a. Add Qumulo Nexus as a service provider by using the SP Entity ID and ACS URL from your Nexus account.

    > **Note**
    > If your IdP is joined to Active Directory (AD), configure the IdP to send the User Principal Name (UPN) as the primary name identifier. This lets the Qumulo cluster use the information provided by the IdP to identify an authenticated user and any remote management actions that she performs on the cluster.

    b. Map the IdP attributes for user email, first name, and last name to the `email`, `firstName`, and `lastName` Nexus attributes.

For more information specific to your SAML IdP, see the following documentation:

- Auth0: Customize SAML Assertions in the Authenticate documentation
- Microsoft Entra/Azure Active Directory: Customize SAML token claims in the Microsoft Entra documentation

> **Note**
> Because Qumulo Nexus SSO integration requires specifying attributes without a namespace, make sure you clear the **Namespace** field when you specify an attribute in Microsoft Entra.

- **Google Workspace:** Set up your own custom SAML app in the Google Workspace Admin Help documentation
- **Okta:** Define attribute statements in the Okta Identity Engine documentation
- **Ping Identity:** Editing an application - SAML in the PingOne documentation

> **Tip**
> If you work together with a storage administrator who manages your organization's Nexus account, provide her with the IdP Metadata URL.

Step 3: Perform Final Single Sign-On (SSO) Configuration in Qumulo Nexus

After you configure the identity provider (IdP), you must perform the final SSO configuration in Qumulo Nexus by using the IdP Metadata URL provided by your system administrator.

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On your organization's page, click **SAML SSO**, and then click **Configure SSO**.
4. On the **Configure SAML SSO**, enter the the **IdP Metadata URL** provided by your system administrator and then click **Complete Configuration**.

   SAML SSO - Enabled is displayed.

Next Steps

After you perform the final SSO configuration, you can click **Users** and then add users to your Nexus account. Every user that you add has SSO enabled by default.

Authorization in Qumulo Nexus

Managing User Access in Qumulo Nexus

This section explains how to manage users and user access by assigning preconfigured user roles in Qumulo Nexus.

How User Roles Work in Qumulo Nexus

Nexus roles are preconfigured for specific user tasks in your Nexus organization.

> **Note**
> You can assign multiple roles to a single user. The permissions granted by these roles are cumulative.

- **View:** Read-only access for viewing cluster information, analytics, and data flow
- **Admin:** Full administrative access to cluster operations, including managing users and the organization
- **Cluster API Access:** For users with configured SSO (page 0) and role-based access control (RBAC) configured for the Qumulo cluster, access to Nexus Remote Management (page 13)

  > **Important**
  > To be able to work with Nexus Remote Management (page 11):
  >
  > - SSO users must have the Cluster API Access role and either the `View` or `Admin` role assigned.
  > - The user's email address in Nexus and Active Directory (AD) for the Qumulo cluster must match. The AD user must also have RBAC configured in Qumulo Core.

- **Cluster Management Admin:** Full ability to create and destroy only Qumulo clusters created by using Nexus

Assigning Roles to Nexus Users

This section explains how to view existing Nexus users, add a Nexus user, and remove a Nexus user.

Prerequisites

- Administrative access to your organization's Nexus account

- Single sign-on (SSO) configured by your organization's storage administrator (page 2) or system administrator (page 5)

To List Existing Users

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On your organization's page, click **Users**

   The **Users** page lists the users in your Nexus organization and the **Full Name**, **Email Address**, and **Role** for each user.
4. To show the roles for a user, click ⋮ > **View User**.

   The **View User Details** lists the user's full name, email address, and **Applied Roles**.

To Add a Nexus User

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On the **Users** page, click **Add Users**.
4. In the **Add Users** side panel, for **Email Address**, enter one or more email addresses.

   > ☑ Tip
   > Separate multiple email addresses with commas (,).
5. (Optional) For SSO users, leave **Do not add as SSO user** unchecked.
6. Under **Roles**, click ⌄ and then select one or more user roles (page 8).
7. Click **Save**.

The **Users** page lists the added users.

To Assign a Nexus Role to a User

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On the **Users** page, next to a user's name, click ⋮ > **Edit User**.
4. In the **Edit User** dialog box, take the following steps:
   a. (Optional) Update the user's **First Name** or **Last Name**.
   b. To add a role, under **Roles** click ⌄ and then select one or more user roles (page 8).

c. To remove a role, under **Roles** click × next to a role

d. Click **Save**.

The **Users** page lists the **Role** assigned to the user.

To Remove a User

> ⚠ Important
>
> - Removing a user from your identity provider (IdP) doesn't revoke user access immediately because existing sessions and access tokens remain active until they expire.
> - To revoke user access immediately, you must remove the user directly from Nexus.

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **Organization Settings**.
3. On the **Users** page, next to a user's name, click ⋮ > **Remove User**.
4. Click **Remove user**.
5. In the **Remove <Username>** dialog box, click **Remove**.

The user is removed from the **Users** page.

Next Steps

After you configure users and assign roles to them in Nexus, you can enable Nexus Remote Support and Nexus Management (page 11) for your Qumulo clusters.

Monitoring and Metrics

Enabling Nexus Remote Support and Nexus Remote Management for Qumulo Core

This section explains how to create a registration key and register your Qumulo cluster from Nexus and how to enable Nexus Remote Support and Nexus Management by using the `qq` CLI.

Qumulo Nexus includes two remote access components:

- For clusters that run Qumulo Core 7.7.1.1 (and higher), Nexus Remote Support lets the Qumulo Care Team access your cluster and work on it remotely.
- For clusters that run Qumulo Core 7.7.3 (and higher), Nexus Remote Management lets users in your organization with single sign-on (SSO) manage your cluster from Nexus.

> ⚠ Important
> To let the Qumulo Care Team provide fast support when you need it most, we strongly recommend enabling both Nexus Remote Support and Nexus Remote Management.

Prerequisites

- Administrative access to your organization's Nexus account
- TCP traffic on port 443 for the `api.nexus.qumulo.com` hostname to allow Nexus connectivity
- For Nexus Remote Management:
  - A configured identity provider (IdP) and enabled SSO for Nexus

    > ☑ Tip
    > For more information, see the storage administrator workflow (page 2) or the system administrator workflow (page 5).

  - A Qumulo cluster joined to Active Directory (AD)
  - Users configured in your IdP with the following roles in Nexus (page 8):
    - `View` or `Admin`
    - `Cluster API Access`
  - Users who log in to Nexus by using the IdP resolve to AD users (page 6).

> **Note**
> The resolved AD user must have appropriate RBAC roles assigned in Qumulo Core.

Step 1: Create a Registration Key in Qumulo Nexus

A registration key associates your Qumulo cluster with your Qumulo Nexus account. It works like a one-time password and is valid for 14 days after being created.

> **Important**
>
> - Nexus displays the registration key only once. Copy this key to a secure location.
> - If you misplace your key before registering your cluster or before it expires:
>
>   1. In the **Manage Registration Keys** section, locate the key, and then click **Revoke**.
>   2. Create a new key.
>
> - Revoking a registration key is a permanent action. However, you can create a new registration key at any time.

1. Log in to Qumulo Nexus.
2. In the upper-right corner, click your username and then click **User Settings**.
3. On the **User Settings** page, click **Registration Keys**.
4. Enter the **Key Name** to identify your registration key and then click **Create Key**.
5. In the **Registration Key Created** dialog box, click **Copy Key** and then click **Done**.

   Your registration key appears in the **Registered Keys** section, which shows the key's name, creation date, and expiration date.

When you use the registration key to register your cluster with Nexus, Nexus removes the key from this the **Registered Keys** list.

Step 2: Register Your Cluster with Nexus

Run the `qq nexus_set_registration` command with the `--join-key` flag and specify your registration key. For example:

```
qq nexus_set_registration \
  --join-key "1A2B3CDEF4"
```

Step 3: Enable Nexus Remote Support and Nexus Remote Management

> ⚠ Important
> To let the Qumulo Care Team provide fast support when you need it most, we strongly recommend enabling both Nexus Remote Support and Nexus Remote Management.

To Enable Nexus Remote Support

For clusters that run Qumulo Core 7.7.1.1 (and higher), Nexus Remote Support lets the Qumulo Care Team access your cluster and work on it remotely.

Run the `qq nexus_set_config` command with the `--enable` and `--enable-remote-support` flags. For example:

```
qq nexus_set_config \
  --enable \
  --enable-remote-support
```

When you enable Nexus Remote Support, Qumulo Core assigns the `qumulosupport:everyone` trustee to the `Qumulo-Support` RBAC role. This RBAC role determines the level of REST API access that Nexus gives to the Qumulo Care Team. For more information, see the Qumulo-Support section in Managing Role-Based Access Control (RBAC) for Users and Groups.

To Enable Nexus Remote Management

For clusters that run Qumulo Core 7.7.3 (and higher), Nexus Remote Management lets users in your organization with single sign-on (SSO) manage your cluster from Nexus.

> ⚠ Important
> Before you begin, review the Nexus Remote Management Prerequisites (page 11).

Run the `qq nexus_set_config` command with the `--enable` and `--enable-sso` flags. For example:

```
qq nexus_set_config \
  --enable \
  --enable-sso
```

Disabling Nexus Functionality

- To unregister your cluster, run the `qq nexus_set_registration` command with the `--delete` flag.
- To disable Nexus Remote Support, run the `qq nexus_set_config` command with the `--disable-remote-support` flag.
- To disable Nexus Remote Management, run the `qq nexus_set_config` command with the `--disable-sso` flag.
- To disable all Nexus functionality, run the `qq nexus_set_config` command with the `--disable` flag.

NeuralProtect in Qumulo Nexus

NeuralProtect in Qumulo Nexus

This section explains Qumulo NeuralProtect in Qumulo Nexus.

NeuralProtect is a threat-detection feature integrated into the Qumulo File System that scans client traffic in near-realtime in order to identify and mitigate malware threats and ransomware attacks before your data is lost.

How NeuralProtect Works

NeuralProtect leverages an industry-leading threat detection engine, integrated into your Qumulo File System, in order to protect your data. NeuralProtect uses multiple detection approaches to identify ransomware, malware, suspicious content, and other threat activity.

- Behavioral ransomware detections identify activity that looks like ransomware encryption, including patterns associated with unknown or zero-day ransomware strains.
- Temporal ransomware detections compare how data changes over time to identify ransomware behavior that might be hidden when looking at only a single point in time.
- Deterministic ransomware detections identify ransomware behavior and known ransomware families by using indicators derived from previously analyzed ransomware strains.
- Insider threat detections identify suspicious encryption activity by users or processes that already have valid access to the data. These detections can help identify smaller, targeted attacks that might not have the volume patterns of broad ransomware activity.
- Malware detections identify known malware and suspicious payloads that might indicate an early-stage compromise before ransomware detonates.

NeuralProtect updates regularly to include the latest known signatures and detection logic.

Protection Defaults and Limits

By default, NeuralProtect enables both quarantine and defensive snapshots.

You can disable quarantine, defensive snapshots, or both in the Nexus UI.

> ⚠ Important
> For best protection, we recommend keeping both quarantine and defensive snapshots enabled and configuring a snapshot policy (page 0).

When quarantine is enabled, NeuralProtect moves detected malware files to the configured quarantine path. The default quarantine path is `/.qumulo_quarantine`.

When defensive snapshots are enabled, NeuralProtect creates snapshots at the root of the file system after a threat detection. A minimum time between defensive snapshots can be configured, and the default is 60 minutes. Defensive snapshots have an expiration time set: snapshots created after encryption detections expire after 7 days, and snapshots created after other threat detections expire after 30 days.

Notes

- NeuralProtect is supported only with clusters connected to Qumulo Nexus (page 11).
- NeuralProtect uses the audit log (page 0) to be notified of write operations. With QFSD 7.9.0 and later, multiple audit destinations are supported, so NeuralProtect can be combined with other uses of the audit function. For earlier QFSD versions, NeuralProtect cannot be combined with other audit log destinations.
- Removing the NeuralProtect's audit log configuration will disable protection.
- In normal operation, the audit log stream to NeuralProtect may drop some events. The number of missed events is extremely small. NeuralProtect periodically runs self-tests to confirm that it is receiving and processing audit events promptly. If NeuralProtect becomes excessively delayed or drops events significantly, it produces an alert. The Qumulo team can help diagnose the issue and restore proper function.
- The NeuralProtect scanning engine runs next to the Qumulo cluster. File data is not sent to the cloud service for scanning nor as a result of detections. To support alerting and support responses, NeuralProtect does send file metadata related to detections to Nexus, including file paths and names.
- For a Cloud Data Fabric (CDF) configuration (page 0), each cluster requires NeuralProtect to be enabled for all new write activity to be scanned.
- To maintain real-time scanning, extremely large files and archives will have some memory-intensive and compute-intensive scanning steps skipped.
- Defensive snapshots occur after NeuralProtect detects a threat or encryption. Because a defensive snapshot can occur after data loss has already begun, we recommend configuring a snapshot policy (page 0) in advance so that you can access a recent version of even the first file attacked.

Next Steps

- To configure NeuralProtect in Nexus, see Configuring NeuralProtect (page 17).

Configuring NeuralProtect in ANQ

This section explains how to enable and configure NeuralProtect on ANQ in Qumulo Nexus.

This section explains how to deploy and configure NeuralProtect for an Azure Native Qumulo (ANQ) deployment from Qumulo Nexus.

Prerequisites

- The `Threat Detection System Configurer` role in Nexus

  A Nexus user with the `Admin` role can grant this role. For more information, see Managing User Access in Qumulo Nexus (page 8).

- ANQ Enterprise
- An ANQ cluster to protect and associated with Qumulo Nexus
- For QFSD versions earlier than 7.9.0, the audit log feature is not in use on the cluster

  > **Note**
  > NeuralProtect uses the audit log configuration for threat detection. With QFSD 7.9.0 and later, multiple audit destinations are supported, so NeuralProtect can be combined with other uses of the audit function. For earlier QFSD versions, NeuralProtect can't share the audit log configuration with any other audit log destination. Replacing the audit log configuration disables protection.

To Enable NeuralProtect

1. Log in to Qumulo Nexus.
2. Go to the ANQ cluster that you want to protect.
3. In the left menu, click **Threat Detection**. If **Threat Detection** is not visible, reach out to the Qumulo Care team for help.
4. Click **Add NeuralProtect**.
5. In the confirmation prompt, click **Yes, Add**.
6. Wait 5 to 10 minutes for the deployment to finish.
7. Click **Go to Dashboard** and confirm that the **NeuralProtect Dashboard** appears and shows that the cluster is protected.

After you enable NeuralProtect, you can return to the Nexus UI at any time to change the protection settings for the cluster.

To Configure Alert Notifications

Nexus users can subscribe to email alerts for the clusters in their account.

1. Go to **User Settings** > **Alerts**.
2. Enable email alerts.
3. Choose the instances that you want to receive alerts from.
4. In the **Alerts to receive** section, under **Threat Detection**, turn on the NeuralProtect alert types that you want.

To Configure Exclude and Suppression Behavior

NeuralProtect lets you configure glob path patterns to control how detection behavior applies to specific parts of the file system.

- In **Threat Detection** > **Settings**, use **Exclude Patterns** to enter comma-separated glob patterns. Exclude patterns fully exclude matching paths from NeuralProtect scanning. Events on matching paths don't trigger scanning at all, so you can't use NeuralProtect events to confirm whether an exclude pattern is being matched.
- In **Threat Detection** > **Suppression Rules**, configure suppression rules for paths where you want NeuralProtect to scan but not send alert notifications. Because scanning still occurs, suppression events can show whether suppression rules are being activated.

In general, we recommend using suppression rules instead of exclude patterns. Suppression rules provide more information about detection behavior and help you confirm that a rule is applied correctly.

NeuralProtect scans are not expected to impact other workloads that use the same files. Scans do not take protocol locks or cause contention that would interfere with other applications.

To Configure Quarantine

When quarantine is enabled, NeuralProtect automatically moves files that contain detected malware to the configured quarantine path.

In **Threat Detection** > **Settings**, you can turn quarantine behavior on or off and configure the **Quarantine path**. The default quarantine path is `/.qumulo_quarantine`.

To Configure Defensive Snapshots

When defensive snapshots are enabled, NeuralProtect creates snapshots at the root of the file system after a threat detection.

In **Threat Detection** > **Settings**, you can turn defensive snapshots on or off and configure the **Minimum interval between snapshots (minutes)**. NeuralProtect will create defensive snapshots no more frequently than this value (default is 60 minutes).